Practice Exam

Exam Content and Format

This AP Computer Science Principles End-of-Course Practice Exam is 2 hours long and consists of 66 multiple-choice questions.

Administering the Practice Exam

This section contains instructions for administering the AP Computer Science Principles Practice Exam. You may wish to use these instructions to create an exam situation that resembles an actual administration. If so, read the indented, boldface directions to the students; all other instructions are for administering the exam and need not be read aloud. Before beginning testing, have all exam materials ready for distribution. These include test booklets and answer sheets.

When you are ready to begin the exam, say:

> **You will be given 2 hours to answer 66 multiple-choice questions. Each question has four answer choices.**
>
> - **For question numbers 1 through 59, mark only the single best answer to each question.**
> - **For the remaining questions, numbered 131 through 137, mark the two best answer choices for each question.**
>
> **Your total score on this multiple-choice exam is based only on the number of questions answered correctly. Points are not deducted for incorrect answers or unanswered questions. When you do not know the answer to a question, you should eliminate as many choices as you can, and then select the best answer among the remaining choices. If you finish before time is called, you may check your work.**
>
> **Programming reference materials are located at the front of the exam. The reference materials provide instructions and explanations to help you understand the format and meaning of the questions on the test. As you encounter programming questions on the test, you should use the reference materials to clarify the behavior of programming statements found in those questions.**
>
> **You have 2 hours for this exam. Open your exam booklet and begin.**

Note Start Time here_____ . Note Stop Time here_____ . After 2 hours, say:

> **Stop working. The exam is over. I will now collect your exam materials.**

Collect an exam booklet and answer sheet from each student.

CollegeBoard AP

Answer Sheet 2018

AP Number Label
(from Student Pack)

AP FPO X123456789 X

AP Exam Label
(from Section I Booklet)

COMPLETE THIS AREA AT EVERY EXAM. **USE NO. 2 PENCIL ONLY**

I am aware of and agree to follow the policies and procedures in the *2017-18 Bulletin for AP Students and Parents* to maintain the security of the exam and the validity of my AP score. I understand and accept that my exam score may be canceled if I do not follow these policies and procedures, if I disregard any exam day instructions, or if my exam is administered improperly. I certify that I am the person whose information appears on this answer sheet.

A. SIGNATURE Sign your legal name as it will appear on your college applications. Date

B. LEGAL NAME Omit apostrophes, Jr., II.

Legal Last Name — First 15 Letters | Legal First Name — First 12 Letters | MI

C. YOUR AP NUMBER

D. EXAM DATE Month Day

E. EXAM START TIME AM PM

F. MULTIPLE-CHOICE BOOKLET SERIAL NUMBER S

G. ONLINE PROVIDER CODE

COMPLETE THIS AREA ONLY ONCE.

I. AREA CODE AND PHONE NUMBER

INTERNATIONAL PHONE

J. SCHOOL YOU ATTEND

SCHOOL CODE | School Name | City | State | Country

K. DATE OF BIRTH

Month: Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec

Day | Year

L. SOCIAL SECURITY NUMBER (Optional)

M. COLLEGE TO RECEIVE YOUR AP SCORE REPORT

COLLEGE CODE

Using the college code listed in the AP Student Pack, indicate the **ONE** college that you want to receive your AP score report.

College Name | City | State | Country

N. CURRENT GRADE LEVEL

Not yet in 9th grade | 9th | 10th | 11th | 12th | No longer in high school

O. STUDENT SEARCH SERVICE®

Colleges and scholarship programs may request your information to inform you of educational opportunities and financial aid.

Would you like us to supply your information?

Yes No

If you don't answer and previously chose to participate in this service, we will continue providing your information.

H. AP EXAM I AM TAKING USING THIS ANSWER SHEET Exam Name: Form: Form Code:

801356

SCHOOL USE ONLY

Section Number: 1 2 3 4 5 6 7 8 9

Fee Reduction Granted: 1 Option 1 2 Option 2

121586-00657 • TF11017E5500 • Printed in U.S.A. Q4546/1-4

PAGE 2

COMPLETE THIS AREA AT EACH EXAM (IF APPLICABLE).

P. SURVEY QUESTIONS — Answer the survey questions in the AP Student Pack. Do not put responses to exam questions in this section.

1 (A)(B)(C)(D)(E)(F)(G)(H)(I)	4 (A)(B)(C)(D)(E)(F)(G)(H)(I)	7 (A)(B)(C)(D)(E)(F)(G)(H)(I)
2 (A)(B)(C)(D)(E)(F)(G)(H)(I)	5 (A)(B)(C)(D)(E)(F)(G)(H)(I)	8 (A)(B)(C)(D)(E)(F)(G)(H)(I)
3 (A)(B)(C)(D)(E)(F)(G)(H)(I)	6 (A)(B)(C)(D)(E)(F)(G)(H)(I)	9 (A)(B)(C)(D)(E)(F)(G)(H)(I)

Q. LANGUAGE — Do not complete this section unless instructed to do so.

If this answer sheet is for the French Language and Culture, German Language and Culture, Italian Language and Culture, Spanish Language and Culture, or Spanish Literature and Culture Exam, please answer the following questions. Your responses will not affect your score.

1. Have you lived or studied for one month or more in a country where the language of the exam you are now taking is spoken?

◯ Yes ◯ No

2. Do you regularly speak or hear the language at home?

◯ Yes ◯ No

QUESTIONS 1–75

Indicate your answers to the exam questions in this section (pages 2 and 3). Mark only one response per question for Questions 1 through 120. If a question has only four answer options, do not mark option E. Answers written in the multiple-choice booklet will not be scored.

COMPLETE MARK ●

EXAMPLES OF INCOMPLETE MARKS

You must use a No. 2 pencil and marks must be complete. Do not use a mechanical pencil. It is very important that you fill in the entire circle darkly and completely. If you change your response, erase as completely as possible. Incomplete marks or erasures may affect your score.

1 (A)(B)(C)(D)(E)	26 (A)(B)(C)(D)(E)	51 (A)(B)(C)(D)(E)
2 (A)(B)(C)(D)(E)	27 (A)(B)(C)(D)(E)	52 (A)(B)(C)(D)(E)
3 (A)(B)(C)(D)(E)	28 (A)(B)(C)(D)(E)	53 (A)(B)(C)(D)(E)
4 (A)(B)(C)(D)(E)	29 (A)(B)(C)(D)(E)	54 (A)(B)(C)(D)(E)
5 (A)(B)(C)(D)(E)	30 (A)(B)(C)(D)(E)	55 (A)(B)(C)(D)(E)
6 (A)(B)(C)(D)(E)	31 (A)(B)(C)(D)(E)	56 (A)(B)(C)(D)(E)
7 (A)(B)(C)(D)(E)	32 (A)(B)(C)(D)(E)	57 (A)(B)(C)(D)(E)
8 (A)(B)(C)(D)(E)	33 (A)(B)(C)(D)(E)	58 (A)(B)(C)(D)(E)
9 (A)(B)(C)(D)(E)	34 (A)(B)(C)(D)(E)	59 (A)(B)(C)(D)(E)
10 (A)(B)(C)(D)(E)	35 (A)(B)(C)(D)(E)	60 (A)(B)(C)(D)(E)
11 (A)(B)(C)(D)(E)	36 (A)(B)(C)(D)(E)	61 (A)(B)(C)(D)(E)
12 (A)(B)(C)(D)(E)	37 (A)(B)(C)(D)(E)	62 (A)(B)(C)(D)(E)
13 (A)(B)(C)(D)(E)	38 (A)(B)(C)(D)(E)	63 (A)(B)(C)(D)(E)
14 (A)(B)(C)(D)(E)	39 (A)(B)(C)(D)(E)	64 (A)(B)(C)(D)(E)
15 (A)(B)(C)(D)(E)	40 (A)(B)(C)(D)(E)	65 (A)(B)(C)(D)(E)
16 (A)(B)(C)(D)(E)	41 (A)(B)(C)(D)(E)	66 (A)(B)(C)(D)(E)
17 (A)(B)(C)(D)(E)	42 (A)(B)(C)(D)(E)	67 (A)(B)(C)(D)(E)
18 (A)(B)(C)(D)(E)	43 (A)(B)(C)(D)(E)	68 (A)(B)(C)(D)(E)
19 (A)(B)(C)(D)(E)	44 (A)(B)(C)(D)(E)	69 (A)(B)(C)(D)(E)
20 (A)(B)(C)(D)(E)	45 (A)(B)(C)(D)(E)	70 (A)(B)(C)(D)(E)
21 (A)(B)(C)(D)(E)	46 (A)(B)(C)(D)(E)	71 (A)(B)(C)(D)(E)
22 (A)(B)(C)(D)(E)	47 (A)(B)(C)(D)(E)	72 (A)(B)(C)(D)(E)
23 (A)(B)(C)(D)(E)	48 (A)(B)(C)(D)(E)	73 (A)(B)(C)(D)(E)
24 (A)(B)(C)(D)(E)	49 (A)(B)(C)(D)(E)	74 (A)(B)(C)(D)(E)
25 (A)(B)(C)(D)(E)	50 (A)(B)(C)(D)(E)	75 (A)(B)(C)(D)(E)

FOR OFFICIAL USE ONLY

Exam		0 1 2 3 4 5 6 7 8 9
		0 1 2 3 4 5 6 7 8 9
Exam		0 1 2 3 4 5 6 7 8 9
		0 1 2 3 4 5 6 7 8 9

SELECTED MEDIA EXAMS	R	W	O	OTHER EXAMS	R	W	O
PT02				TOTAL			
PT03				Subscore (if applicable)			
PT04				Subscore (if applicable)			

DO NOT WRITE IN THIS AREA

QUESTIONS 76–120

Be sure each mark is dark and completely fills the circle. If a question has only four answer options, do not mark option E.

76	A B C D E	91	A B C D E	106	A B C D E
77	A B C D E	92	A B C D E	107	A B C D E
78	A B C D E	93	A B C D E	108	A B C D E
79	A B C D E	94	A B C D E	109	A B C D E
80	A B C D E	95	A B C D E	110	A B C D E
81	A B C D E	96	A B C D E	111	A B C D E
82	A B C D E	97	A B C D E	112	A B C D E
83	A B C D E	98	A B C D E	113	A B C D E
84	A B C D E	99	A B C D E	114	A B C D E
85	A B C D E	100	A B C D E	115	A B C D E
86	A B C D E	101	A B C D E	116	A B C D E
87	A B C D E	102	A B C D E	117	A B C D E
88	A B C D E	103	A B C D E	118	A B C D E
89	A B C D E	104	A B C D E	119	A B C D E
90	A B C D E	105	A B C D E	120	A B C D E

QUESTIONS 121–126

For Students Taking AP Biology

Write your answer in the boxes at the top of the griddable area and fill in the corresponding circles. Mark only one circle in any column. You will receive credit only if the circles are filled in correctly.

121 122 123 124 125 126

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DO NOT WRITE IN THIS AREA

QUESTIONS 131–142

For Students Taking AP Computer Science Principles, AP Physics 1, or AP Physics 2

Mark two responses per question. You will receive credit only if both correct responses are selected.

131	A B C D	135	A B C D	139	A B C D
132	A B C D	136	A B C D	140	A B C D
133	A B C D	137	A B C D	141	A B C D
134	A B C D	138	A B C D	142	A B C D

COMPLETE THIS AREA ONLY ONCE.

PAGE 4

R. YOUR MAILING ADDRESS Use the address abbreviations from your AP Student Pack. Fill in only one circle per column. Indicate a space in your address by leaving a blank box; do not grid that column.

STREET ADDRESS (include street number, street name, apartment number, etc.)

CITY

ZIP OR POSTAL CODE

COUNTRY CODE

STATE

AK HI MI NY VT
AL IA MN OH WA
AR ID MO OK WI
AZ IL MS OR WV
CA IN MT PA WY
CO KS NC RI Puerto Rico
CT KY ND SC
DC LA NE SD AA
DE MA NH TN AE
FL MD NJ TX AP
GA ME NM UT Other
NV VA

S. FOR STUDENTS OUTSIDE THE UNITED STATES ONLY If your address does not fit in the spaces provided in Item R, fill in as many circles as you can, then fill in the circle in Item S and print the remainder of your address in the spaces provided.

Address | City | State or Province | Country | ZIP or Postal Code

U. EMAIL ADDRESS By providing your email address, you are granting the College Board permission to use your email address in accordance with the policies in the *2017-18 Bulletin for AP Students and Parents*.

T. STUDENT IDENTIFIER (Student ID Number)

V. SEX

- Female
- Male

W. WHICH LANGUAGE DO YOU KNOW BEST?

- English
- English and another language about the same
- Another language

X. RACIAL/ETHNIC GROUP

Please answer both questions about Hispanic origin and about race. For the following questions about your identity, Hispanic origins are not races.

(You may mark all that apply.)

a. Are you of Hispanic, Latino, or Spanish origin?

- No, not of Hispanic, Latino, or Spanish origin
- Yes, Cuban
- Yes, Mexican
- Yes, Puerto Rican
- Yes, another Hispanic, Latino, or Spanish origin

b. What is your race?

- American Indian or Alaska Native
- Asian (including Indian subcontinent and Philippines origin)
- Black or African American (including Africa and Afro-Caribbean origin)
- Native Hawaiian or other Pacific Islander
- White (including Middle Eastern origin)

Y. PARENTAL EDUCATION LEVEL

In the first column, indicate the highest level of education of one parent/guardian, and indicate whether this is your mother/female guardian or father/male guardian. Then, if applicable, indicate the highest level of education of your other parent/guardian in the second column, and indicate whether this is your mother/female guardian or father/male guardian.

- **Mother or female guardian**
- **Father or male guardian**

- Grade school
- Some high school
- High school diploma or equivalent
- Vocational or trade school
- Some college
- Associate or two-year degree
- Bachelor's or four-year degree
- Some graduate or professional school
- Graduate or professional degree

AP® Computer Science Principles Exam

Multiple Choice

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

At a Glance

Total Time
2 hours
Number of Questions
66
Percent of Total Score
60%
Writing Instrument
Pencil required
Electronic Device
None allowed

Instructions

Exam reference materials are located at the front of this booklet.

This practice exam contains 66 multiple-choice questions. Indicate all of your answers to the multiple-choice questions on the answer sheet. No credit will be given for anything written in this exam booklet, but you may use the booklet for notes or scratch work.

For questions 1 through 59, select the single best answer choice for each question. After you have decided which of the choices is best, fill in the appropriate letter in the corresponding space on the answer sheet.

For questions 131 through 137, select the two best answer choices for each question. After you have decided which two of the choices are best, enter both letters in the corresponding space on the answer sheet.

Use your time effectively, working as quickly as you can without losing accuracy. Do not spend too much time on any one question. Go on to other questions and come back to the ones you have not answered if you have time. It is not expected that everyone will know the answers to all of the multiple-choice questions.

Your total score on the exam is based only on the number of questions answered correctly. Points are not deducted for incorrect answers or unanswered questions.

COMPUTER SCIENCE PRINCIPLES

Time—2 Hours

Number of Questions—66

Programming reference materials are included on the following pages. As AP Computer Science Principles does not designate any particular programming language, these reference materials provide instructions and explanations to help you understand the format and meaning of the questions you will see on the exam. The reference materials include two programming formats, text-based and block-based.

Programming instructions use four data types: numbers, Booleans, strings, and lists.

Instructions from any of the following categories may appear on the exam:

- Assignment, Display, and Input
- Arithmetic Operators and Numeric Procedures
- Relational and Boolean Operators
- Selection
- Iteration
- List Operations
- Procedures
- Robot

GO ON TO THE NEXT PAGE.

REFERENCE MATERIALS

Instruction	Explanation
Assignment, Display, and Input	
Text: `a ← expression` Block: `a ← expression`	Evaluates `expression` and assigns the result to the variable `a`.

Text:
`DISPLAY (expression)`
 	
Block:
`DISPLAY expression`
	Displays the value of `expression`, followed by a space.

Text:
`INPUT ()`
 	
Block:
`INPUT`
	Accepts a value from the user and returns it.

Arithmetic Operators and Numeric Procedures	

Text and Block:
`a + b`
`a - b`
`a * b`
`a / b`
	The arithmetic operators `+`, `-`, `*`, and `/` are used to perform arithmetic on `a` and `b`.
	
	For example, `3 / 2` evaluates to `1.5`.

Text and Block: `a MOD b`	Evaluates to the remainder when `a` is divided by `b`. Assume that `a` and `b` are positive integers. For example, `17 MOD 5` evaluates to `2`.

Text:
`RANDOM (a, b)`
 	
Block:
`RANDOM a, b`
	Evaluates to a random integer from `a` to `b`, including `a` and `b`.
	
	For example, `RANDOM (1, 3)` could evaluate to `1`, `2`, or `3`.

Relational and Boolean Operators	

Text and Block:
`a = b`
`a ≠ b`
`a > b`
`a < b`
`a ≥ b`
`a ≤ b`
	The relational operators `=`, `≠`, `>`, `<`, `≥`, and `≤` are used to test the relationship between two variables, expressions, or values.
	
	For example, `a = b` evaluates to `true` if `a` and `b` are equal; otherwise it evaluates to `false`.

Text: `NOT condition` Block: `NOT condition`	Evaluates to `true` if condition is `false`; otherwise evaluates to `false`.
Text: `condition1 AND condition2` Block: `condition1 AND condition2`	Evaluates to `true` if both `condition1` and `condition2` are `true`; otherwise evaluates to `false`.

REFERENCE MATERIALS

Instruction	Explanation
Relational and Boolean Operators (continued)	

Text:
`condition1 OR condition2`
 	
Block:
`condition1` `OR` `condition2`
	Evaluates to `true` if `condition1` is `true` or if `condition2` is `true` or if both `condition1` and `condition2` are `true`; otherwise evaluates to `false`.

Selection	

Text:
`IF (condition)`
`{`
`  <block of statements>`
`}`
 	
Block:
`IF condition`
`  block of statements`
	The code in `block of statements` is executed if the Boolean expression `condition` evaluates to `true`; no action is taken if `condition` evaluates to `false`.

Text:
`IF (condition)`
`{`
`  <first block of statements>`
`}`
`ELSE`
`{`
`  <second block of statements>`
`}`
 	
Block:
`IF condition`
`  first block of statements`
`ELSE`
`  second block of statements`
	The code in `first block of statements` is executed if the Boolean expression `condition` evaluates to `true`; otherwise the code in `second block of statements` is executed.

Iteration	

Text:
`REPEAT n TIMES`
`{`
`  <block of statements>`
`}`
 	
Block:
`REPEAT n TIMES`
`  block of statements`
	The code in `block of statements` is executed `n` times.

Instruction	Explanation
Iteration (continued)	

Text:
`REPEAT UNTIL (condition)`
`{`
`  <block of statements>`
`}`
 	
Block:
`REPEAT UNTIL condition`
`  block of statements`
	The code in `block of statements` is repeated until the Boolean expression `condition` evaluates to `true`.

List Operations	
For all list operations, if a list index is less than 1 or greater than the length of the list, an error message is produced and the program terminates.	

Text:
`list[i]`
 	
Block:
`list i`
	Refers to the element of `list` at index `i`. The first element of `list` is at index `1`.

Text:
`list[i] ← list[j]`
 	
Block:
`list i ← list j`
	Assigns the value of `list[j]` to `list[i]`.

Text:
`list ← [value1, value2, value3]`
 	
Block:
`list ← value1, value2, value3`
	Assigns `value1`, `value2`, and `value3` to `list[1]`, `list[2]`, and `list[3]`, respectively.

Text:
`FOR EACH item IN list`
`{`
`  <block of statements>`
`}`
 	
Block:
`FOR EACH item IN list`
`  block of statements`
	The variable `item` is assigned the value of each element of `list` sequentially, in order from the first element to the last element. The code in `block of statements` is executed once for each assignment of `item`.

REFERENCE MATERIALS

Instruction	Explanation
List Operations (continued)	
Text: `INSERT (list, i, value)` Block: `INSERT` [`list, i, value`]	Any values in `list` at indices greater than or equal to `i` are shifted to the right. The length of list is increased by 1, and `value` is placed at index `i` in `list`.
Text: `APPEND (list, value)` Block: `APPEND` [`list, value`]	The length of `list` is increased by 1, and `value` is placed at the end of `list`.
Text: `REMOVE (list, i)` Block: `REMOVE` [`list, i`]	Removes the item at index `i` in `list` and shifts to the left any values at indices greater than `i`. The length of `list` is decreased by 1.
Text: `LENGTH (list)` Block: `LENGTH` [`list`]	Evaluates to the number of elements in `list`.
Procedures	

Text:
`PROCEDURE name (parameter1,`
`parameter2, ...)`
`{`
`<instructions>`
`}`
 	
Block:
`PROCEDURE name` [`parameter1,`
`parameter2,...`]
[`instructions`]
	A procedure, `name`, takes zero or more parameters. The procedure contains programming instructions.

Text:
`PROCEDURE name (parameter1,`
`parameter2, ...)`
`{`
`<instructions>`
`RETURN (expression)`
`}`
 	
Block:
`PROCEDURE name` [`parameter1,`
`parameter2,...`]
[`instructions`]
`RETURN` [`expression`]
	A procedure, `name`, takes zero or more parameters. The procedure contains programming instructions and returns the value of `expression`. The `RETURN` statement may appear at any point inside the procedure and causes an immediate return from the procedure back to the calling program.

REFERENCE MATERIALS

Instruction	Explanation
Robot	
If the robot attempts to move to a square that is not open or is beyond the edge of the grid, the robot will stay in its current location and the program will terminate.	

Text:
`MOVE_FORWARD ()`
 	
Block:
`MOVE_FORWARD`
	The robot moves one square forward in the direction it is facing.

Text:
`ROTATE_LEFT ()`
 	
Block:
`ROTATE_LEFT`
	The robot rotates in place 90 degrees counterclockwise (i.e., makes an in-place left turn).

Text:
`ROTATE_RIGHT ()`
 	
Block:
`ROTATE_RIGHT`
	The robot rotates in place 90 degrees clockwise (i.e., makes an in-place right turn).

Text:
`CAN_MOVE (direction)`
 	
Block:
`CAN_MOVE` `direction`
	Evaluates to `true` if there is an open square one square in the direction relative to where the robot is facing; otherwise evaluates to `false`. The value of direction can be `left`, `right`, `forward`, or `backward`.

Directions: Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

1. The code segment below is intended to swap the values of the variables `first` and `second` using a temporary variable, `temp`.

```
temp ← first
first ← second
<MISSING CODE>
```

Which of the following can be used to replace `<MISSING CODE>` so that the code segment works as intended?

(A) `second ← first`

(B) `second ← temp`

(C) `temp ← first`

(D) `temp ← second`

---

2. A bank customer receives an e-mail from a sender claiming to be a bank employee. The e-mail asks the customer to provide personal information and to call a phone number if he or she has any questions. The customer suspects the e-mail might be a phishing attempt. Which of the following responses is most likely to be a privacy risk for the bank customer?

(A) Calling the bank at its official phone number to ask whether the request for personal information is legitimate

(B) Calling the phone number given in the e-mail and providing the personal information over the phone

(C) Checking that the domain name of the sender's e-mail address is associated with the bank

(D) Conducting a Web search to see if other people have received similar requests for personal information

GO ON TO THE NEXT PAGE.

3. Which of the following would be the best use of citizen science?

(A) An experiment that requires all participants to be working in the same laboratory

(B) An experiment that requires expensive equipment to conduct

(C) An experiment that requires data measurements to be taken in many different locations

(D) An experiment that requires specialized knowledge and training to conduct

---

4. A student is writing a program to model different real-world events using simulations. Which of the following simulations will generate a result that would best be stored using a Boolean variable?

(A) A simulation of flipping a fair coin

(B) A simulation of rolling a fair die (with sides numbered 1 through 6)

(C) A simulation of the temperature in a location over time

(D) A simulation of traffic patterns on a road

GO ON TO THE NEXT PAGE.

5. The ticket prices at a movie theater are given below.

Type of Ticket	Price (in dollars)
Regular	12
Child (ages 12 and below)	9
Senior (ages 60 and above)	9

Additional $5 fee for 3-D movies

A programmer is creating an algorithm to set the value of `ticketPrice` based on the information in the table. The programmer uses the integer variable `age` for the age of the moviegoer. The Boolean variable `is3D` is `true` when the movie is 3-D and `false` otherwise.

Which of the following code segments correctly sets the value of `ticketPrice` ?

(A)

```
ticketPrice ← 12
IF (age ≤ 12) OR (age ≥ 60)
    ticketPrice ← 9
IF (is3D)
    ticketPrice ← 17
```

(B)

```
ticketPrice ← 12
IF (age ≤ 12) OR (age ≥ 60)
    ticketPrice ← 9
ELSE
    ticketPrice ← 17
```

(C)

```
ticketPrice ← 12
IF (age ≤ 12) OR (age ≥ 60)
    ticketPrice ← 9
IF (is3D)
    ticketPrice ← ticketPrice + 5
```

(D)

```
ticketPrice ← 12
IF (age ≤ 12) OR (age ≥ 60)
    ticketPrice ← 9
ELSE
    ticketPrice ← ticketPrice + 5
```

GO ON TO THE NEXT PAGE.

6. Which of the following best describes the role of the Internet Engineering Task Force (IETF) ?

(A) Developing standards and protocols for Internet communication

(B) Preventing copyrighted materials from being illegally distributed online

(C) Preventing malicious software from being distributed online

(D) Verifying the ownership of encrypted keys used in secured messages

---

7. Consider the following program, which uses the variables `start`, `end`, and `current`.

```
start ← 1
end ← 20
current ← 3
start ← current
current ← current + 1
DISPLAY (start)
DISPLAY (current)
```

What is displayed as a result of executing the program?

(A) `1 3`

(B) `3 3`

(C) `3 4`

(D) `4 4`

GO ON TO THE NEXT PAGE.

8. A smartphone stores the following data for each photo that is taken using the phone.

- The file name of the photo
- The date and time the photo was taken
- The geographic location where the photo was taken

Assume that all of the photos that have been taken on the phone are accessible. Which of the following can be determined using the photo data described above?

I. The number of photos that were taken at a particular geographic location

II. The number of photos that were taken in the last year

III. The name of the person who took the most recent photo

(A) III only

(B) I and II only

(C) I and III only

(D) I, II, and III

GO ON TO THE NEXT PAGE.

9. Which of the following best describes one of the benefits of using an iterative and incremental process of program development?

(A) It allows programmers to implement algorithmic solutions to otherwise unsolvable problems.

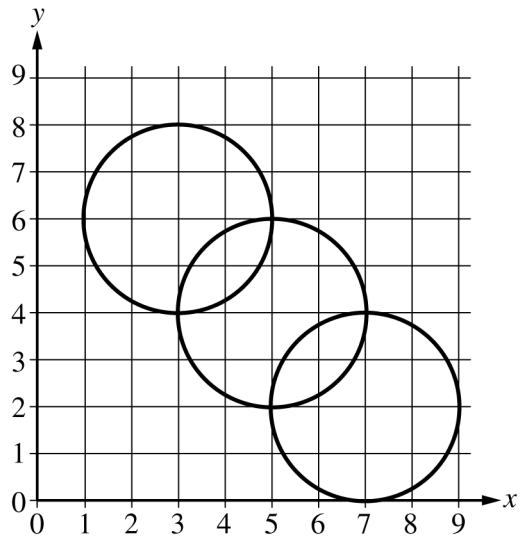
(B) It eliminates the need for programmers to test completed programs.

(C) It enables programmers to create programs that use the lowest-level abstractions available.

(D) It helps programmers identify errors as components are added to a working program.

GO ON TO THE NEXT PAGE.

10. The procedure `DrawCircle (x, y, r)` can be used to draw a circle on a coordinate grid. The circle is centered at the coordinate `(x, y)` and has a radius of `r` units. The procedure will be used to draw the following figure on a coordinate grid.



Which of the following code segments can be used to draw the figure?

(A)
```
xPos ← 3
yPos ← 6
REPEAT 3 TIMES
{
   DrawCircle (xPos, yPos, 2)
   xPos ← xPos + 2
   yPos ← yPos + 2
}
```

(B)
```
xPos ← 3
yPos ← 6
REPEAT 3 TIMES
{
   DrawCircle (xPos, yPos, 2)
   xPos ← xPos + 2
   yPos ← yPos - 2
}
```

(C)
```
xPos ← 7
yPos ← 2
REPEAT 3 TIMES
{
   DrawCircle (xPos, yPos, 2)
   xPos ← xPos + 2
   yPos ← yPos + 2
}
```

(D)
```
xPos ← 7
yPos ← 2
REPEAT 3 TIMES
{
   DrawCircle (xPos, yPos, 2)
   xPos ← xPos + 2
   yPos ← yPos - 2
}
```

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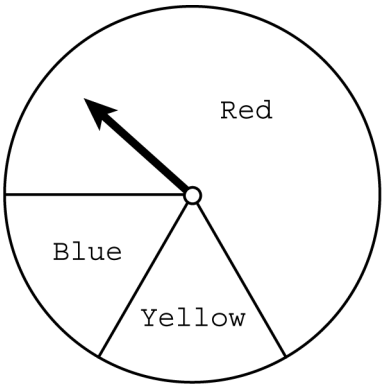
11. A spinner is divided into three sections. The sector labeled `Red` is four times as large as each of the sectors labeled `Blue` and `Yellow`, which are of equal size.



The procedure below is intended to simulate the behavior of the spinner.

```
PROCEDURE Spinner
    <MISSING CODE>
```

Which of the following can be used to replace `<MISSING CODE>` so that the procedure correctly simulates the spinner?

(A)

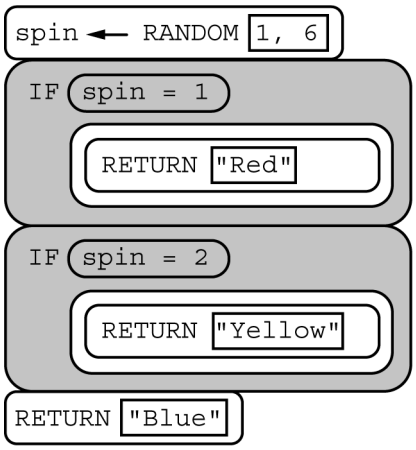
```
spin ← RANDOM [1, 3]
IF (spin = 1)
    RETURN ["Red"]
IF (spin = 2)
    RETURN ["Yellow"]
RETURN ["Blue"]
```

(B)

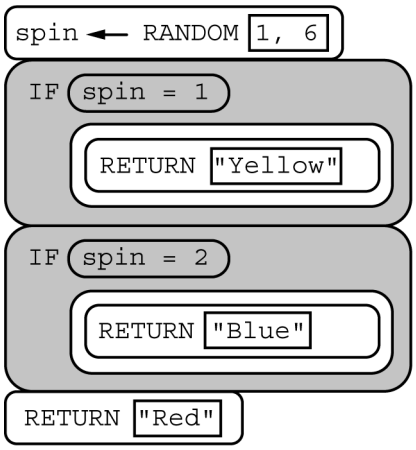
```
spin ← RANDOM [1, 3]
IF (spin = 1)
    RETURN ["Yellow"]
IF (spin = 2)
    RETURN ["Blue"]
RETURN ["Red"]
```

GO ON TO THE NEXT PAGE.

(C)



(D)



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Questions 12–13 refer to the information below.

A social media site allows users to send messages to each other. A group of researchers gathered user data for the first 10 years of the site's existence. Some of the data are summarized in the table below, along with some of the company milestones.

Year	Milestone	Total Number of Registered Users (in millions)	Average Number of Active Daily Users (in millions)	Average Number of Daily Messages Sent Per User	Average Number of Characters Per Message
1	Web site launched	25.4	0.8	3.6	360
2		26.0	0.7	3.5	362
3		26.5	0.6	3.5	358
4		26.9	0.6	3.4	360
5	Mobile app released	27.4	0.9	3.3	269
6		28.0	0.9	3.4	242
7		28.6	1.1	3.5	195
8		29.1	1.2	3.5	176
9		29.6	1.1	3.6	104
10		30.2	1.1	3.6	96

12. The researchers noticed that the total number of registered users appears to be increasing at about a constant rate. If this pattern continues, which of the following best approximates the total number of registered users, in millions, in year 12 (two years after the last entry in the table) ?

(A) 30.6

(B) 31.2

(C) 31.8

(D) 32.4

13. Which of the following hypotheses is most consistent with the data in the table?

(A) The mobile app release did not have any effect on the average number of daily messages sent per user.

(B) The mobile app release discouraged new user registration on the site.

(C) The mobile app release led to users being less frequently active on the site.

(D) The mobile app release led to users tending to write shorter messages.

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14. Consider the two programs below.

Program A:

```
i ← 1
REPEAT UNTIL i > 10
    DISPLAY i
    i ← i + 1
```

Program B:

```
i ← 1
REPEAT UNTIL i > 10
    i ← i + 1
    DISPLAY i
```

Which of the following best compares the values displayed by programs A and B?

(A) Program A and program B display identical values.

(B) Program A and program B display the same values in different orders.

(C) Program A and program B display the same number of values, but the values differ.

(D) Program A and program B display a different number of values.

---

15. Which of the following is a true statement about Internet communication?

(A) Devices from different manufacturers are required to run the same operating system to communicate overthe Internet.

(B) Every device connected to the Internet is assigned a digital certificate by a certificate authority.

(C) Every device connected to the Internet is assigned an Internet protocol (IP) address.

(D) Every device connected to the Internet requires a high-bandwidth connection to enable redundant routing to each device.

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16. A user reads reviews of a popular artist's new album and then downloads the album from the Web site of a licensed online music seller. Which of the following is LEAST likely to be a consequence of this action?

(A) Advertisements for the artist's other albums will be displayed when the user visits a different Web site.

(B) Album reviews from other people who live nearby will be displayed to the user.

(C) Similar artists will be recommended to the user based on the user's download selection.

(D) The user will be in violation of the Digital Millennium Copyright Act (DMCA).

---

17. Which of the following actions could be used to help reduce the digital divide?

I. Providing free education and training on how to use computing devices

II. Providing free or low-cost computing devices to low-income individuals

III. Providing networks and infrastructure to people in remote areas

A) III only

B) I and II only

C) II and III only

D) I, II, and III

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18. A population researcher is interested in predicting the number of births that will occur in a particular community. She created a computer model that uses data from the past ten years, including number of residents and the number of babies born. The model predicted that there would be 200 births last year, but the actual number of births last year was only 120. Which of the following strategies is LEAST likely to provide a more accurate prediction?

(A) Gathering data for additional years to try to identify patterns in birth rates

(B) Refining the model used in the computer simulation to more closely reflect the data from the past ten years

(C) Removing as many details from the model as possible so that calculations can be performed quickly

(D) Taking into consideration more information about the community, such as the ages of residents

---

19. A library of e-books contains metadata for each book. The metadata are intended to help a search feature find books that users are interested in. Which of the following is LEAST likely to be contained in the metadata of each e-book?

(A) An archive containing previous versions of the e-book

(B) The author and title of the e-book

(C) The date the e-book was first published

(D) The genre of the e-book (e.g., comedy, fantasy, romance, etc.)

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20. An application program interface (API) provides a procedure `Max`, which returns the greater of its two integer arguments.

A programmer would like to find the greatest of three integer values `a`, `b`, and `c`. Which of the following expressions will produce the desired result in every case?

(A) `Max (Max (a, b), c)`

(B) `Max (a, b) - Max (b, c)`

(C) `Max (a, b) + Max (b, c) – Max (a, c)`

(D) `(Max (a, b) + Max (b, c)) / 2`

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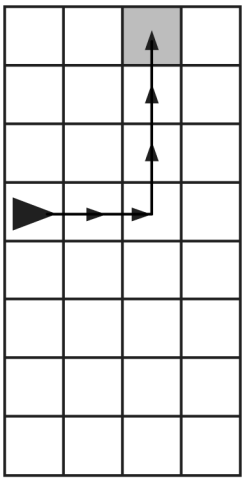
21. The following question uses a robot in a grid of squares. The robot is represented by a triangle, which is initially facing right.



Consider the procedures below.

```
PROCEDURE MoveXTimes [x]
    REPEAT x TIMES
        MOVE_FORWARD
```

```
PROCEDURE RightXTimes [x]
    REPEAT x TIMES
        ROTATE_RIGHT
```

Which of the following code segments will move the robot to the gray square?

(A)

```
MoveXTimes [2]
RightXTimes [1]
MoveXTimes [3]
```

(B)

```
MoveXTimes [2]
RightXTimes [3]
MoveXTimes [3]
```

(C)

```
MoveXTimes [3]
RightXTimes [1]
MoveXTimes [3]
```

(D)

```
MoveXTimes [3]
RightXTimes [3]
MoveXTimes [3]
```

GO ON TO THE NEXT PAGE.

22. A student is creating a procedure to determine whether the weather for a particular month was considered very hot. The procedure takes as input a list containing daily high temperatures for a particular month. The procedure is intended to return `true` if the daily high temperature was at least 90 degrees for a majority of days in the month and return `false` otherwise.

```
PROCEDURE IsHot (temperatureList)
{
   total ← 0
   counter ← 0
   FOR EACH temperature IN temperatureList
   {
      IF (temperature ≥ 90)
      {
         counter ← counter + 1
      }
      total ← total + 1
   }
   RETURN (<MISSING CODE>)
}
```

Which of the following can be used to replace `<MISSING CODE>` so that the procedure works as intended?

(A) `counter < 0.5 * total`

(B) `counter > 0.5 * total`

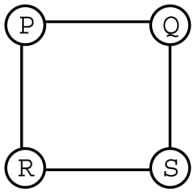
(C) `total < 0.5 * counter`

(D) `total > 0.5 * counter`

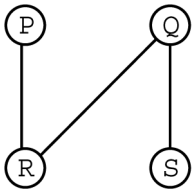
GO ON TO THE NEXT PAGE.

23. The following figures represent different ways of configuring a network of physically linked computers labeled P, Q, R, and S. A line between two computers indicates that the computers can communicate directly with each other. In which configuration is it NOT possible to have redundant routing between computers P and S?

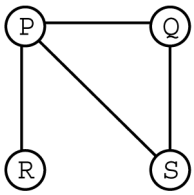
(A)



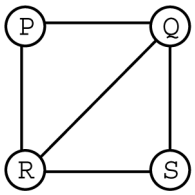
(B)



(C)



(D)



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Questions 24–25 refer to the information below.

Byte pair encoding is a data encoding technique. The encoding algorithm looks for pairs of characters that appear in the string more than once and replaces each instance of that pair with a corresponding character that does not appear in the string. The algorithm saves a list containing the mapping of character pairs to their corresponding replacement characters.

For example, the string `"THIS_IS_THE_BEST_WISH"` can be encoded as `"%#_#_%E_BEST_W#H"` by replacing all instances of `"TH"` with `"%"` and replacing all instances of `"IS"` with `"#"`.

24. Which of the following statements about byte pair encoding is true?

    (A) Byte pair encoding is an example of a lossy transformation because it discards some of the data in the original string.

    (B) Byte pair encoding is an example of a lossy transformation because some pairs of characters are replaced by a single character.

    (C) Byte pair encoding is an example of a lossless transformation because an encoded string can be restored to its original version.

    (D) Byte pair encoding is an example of a lossless transformation because it can be used to transmit messages securely.

---

25. For which of the following strings is it NOT possible to use byte pair encoding to shorten the string's length?

    (A) `"BANANA"`

    (B) `"LEVEL_UP"`

    (C) `"MEET_ME_LATER"`

    (D) `"NEITHER_HERE_NOR_THERE"`

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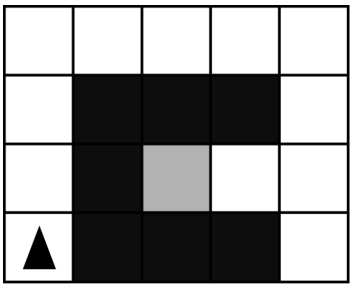
26. The grid below contains a robot represented as a triangle, initially facing up. The robot can move into a white or gray square but cannot move into a black region.



The code segment below uses the procedure `GoalReached`, which evaluates to `true` if the robot is in the gray square and evaluates to `false` otherwise.

```
REPEAT UNTIL (GoalReached ())
{
   <MISSING CODE>
}
```

Which of the following replacements for `<MISSING CODE>` can be used to move the robot to the gray square?

(A)
```
IF (CAN_MOVE (right))
{
   ROTATE_RIGHT ()
}
MOVE_FORWARD ()
```

(B)
```
IF (CAN_MOVE (right))
{
   ROTATE_RIGHT ()
   MOVE_FORWARD ()
}
```

(C)
```
IF (CAN_MOVE (forward))
{
   MOVE_FORWARD ()
}
ROTATE_RIGHT ()
```

(D)
```
IF (CAN_MOVE (forward))
{
   MOVE_FORWARD ()
   ROTATE_RIGHT ()
}
```

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27. A student wrote the following code for a guessing game.

```
Line 1:  secretNumber ← RANDOM (1, 100)
Line 2:  win ← false
Line 3:  REPEAT UNTIL (win)
Line 4:  {
Line 5:     DISPLAY ("Guess a number.")
Line 6:     guess ← INPUT ()
Line 7:     IF (guess = secretNumber)
Line 8:     {
Line 9:        DISPLAY ("You got it right!")
Line 10:    }
Line 11:    ELSE
Line 12:    {
Line 13:       IF (guess > secretNumber)
Line 14:       {
Line 15:          DISPLAY ("Your guess is too high.")
Line 16:       }
Line 17:       ELSE
Line 18:       {
Line 19:          DISPLAY ("Your guess is too low.")
Line 20:       }
Line 21:    }
Line 22: }
```

While debugging the code, the student realizes that the loop never terminates. The student plans to insert the instruction `win ← true` somewhere in the code. Where could `win ← true` be inserted so that the code segment works as intended?

(A) Between line 6 and line 7

(B) Between line 9 and line 10

(C) Between line 20 and 21

(D) Between line 21 and 22

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28. A text-editing application uses binary sequences to represent each of 200 different characters. What is the minimum number of bits needed to assign a unique bit sequence to each of the possible characters?

(A) 4

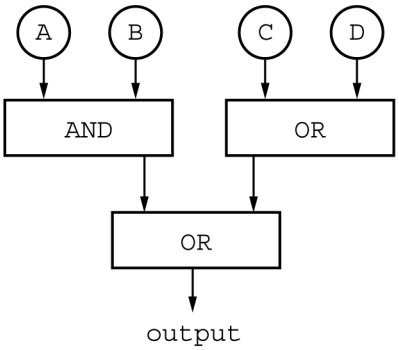
(B) 6

(C) 7

(D) 8

---

29. The diagram below shows a circuit composed of three logic gates. Each gate takes two inputs and produces a single output.



For which of the following input values will the circuit have an output of `false`?

(A) `A = true, B = false, C = false, D = false`

(B) `A = true, B = true, C = false, D = false`

(C) `A = false, B = false, C = true, D = true`

(D) `A = false, B = false, C = false, D = true`

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30. A video-streaming service maintains a database of information about its customers and the videos they have watched.

The program below analyzes the data in the database and compares the number of viewers of science fiction videos to the number of viewers of videos of other genres. It uses the procedure `Analysis (category)`, which returns the number of unique users who viewed videos of a given category in the past year. The `Analysis` procedure takes approximately 1 hour to return a result, regardless of the number of videos of the given genre. All other operations happen nearly instantaneously.

```
sciFiFans ← Analysis ("science fiction")
genreList ← ["comedy", "drama", "mystery", "romance"]
FOR EACH genre IN genreList
{
   IF (Analysis (genre) > sciFiFans)
   {
      DISPLAY (genre)
   }
}
```

Which of the following best approximates the amount of time it takes the program to execute?

(A) 1 hour

(B) 2 hours

(C) 4 hours

(D) 5 hours

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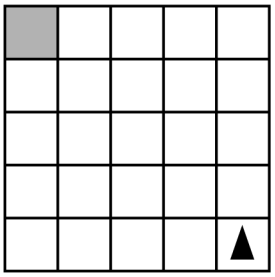
31. The question below uses a robot in a grid of squares. The robot is represented as a triangle, which is initially in the bottom right square of the grid and facing toward the top of the grid.



The following programs are each intended to move the robot to the gray square. Program II uses the procedure `GoalReached`, which returns `true` if the robot is in the gray square and returns `false` otherwise.

Program I:

```
REPEAT 4 TIMES
    MOVE_FORWARD
    ROTATE_LEFT
    MOVE_FORWARD
    ROTATE_RIGHT
```

Program II:

```
REPEAT UNTIL GoalReached
    IF CAN_MOVE forward
        MOVE_FORWARD
    ELSE
        ROTATE_LEFT
```

Which of the following statements is true?

(A) Program I correctly moves the robot to the gray square, but program II does not.

(B) Program II correctly moves the robot to the gray square, but program I does not.

(C) Both program I and program II correctly move the robot to the gray square.

(D) Neither program I nor program II correctly moves the robot to the gray square.

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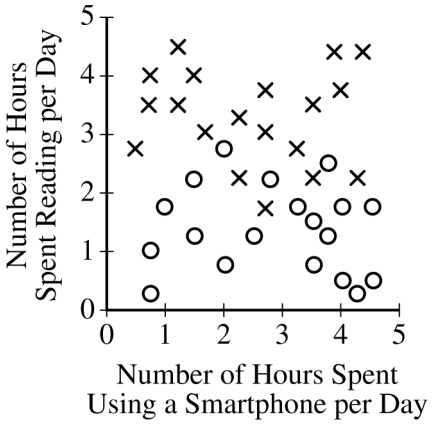
32. Participants in a survey were asked how many hours per day they spend reading, how many hours per day they spend using a smartphone, and whether or not they would be interested in a smartphone application that lets users share book reviews.

The data from the survey are represented in the graph below. Each × represents a survey participant who said he or she was interested in the application, and each ○ represents a participant who said he or she was not interested.



Which of the following hypotheses is most consistent with the data in the graph?

(A) Participants who read more were generally more likely to say they are interested in the application.

(B) Participants who read more were generally less likely to say they are interested in the application.

(C) Participants who use a smartphone more were generally more likely to say they read more.

(D) Participants who use a smartphone more were generally less likely to say they read more.

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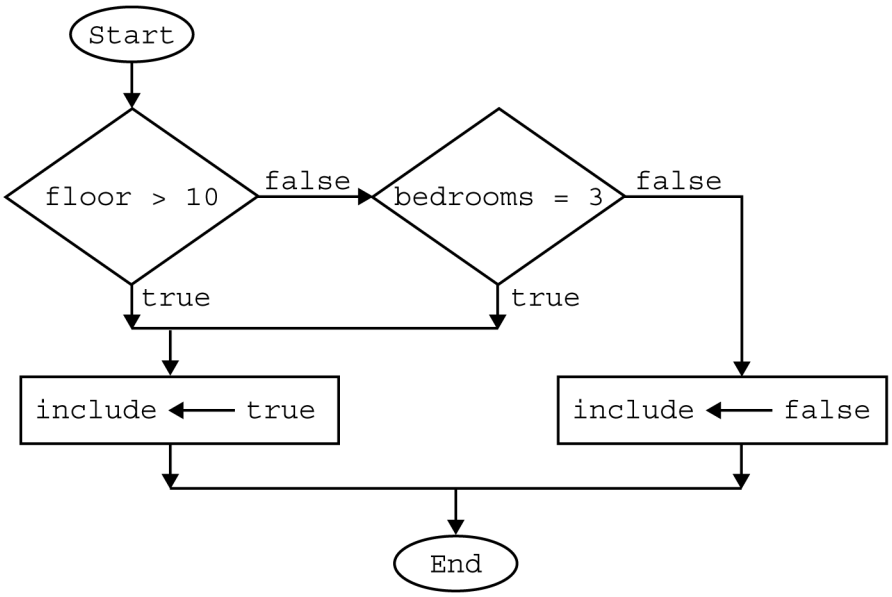
33. A flowchart is a way to visually represent an algorithm. The flowchart below is used by an apartment rental Web site to set the variable `include` to `true` for apartments that meet certain criteria.

Block	Explanation
Oval	The start or end of the algorithm
Diamond	A conditional or decision step, where execution proceeds to the side labeled `true` if the condition is true and to the side labeled `false` otherwise
Rectangle	One or more processing steps, such as a statement that assigns a value to a variable



Which of the following statements is equivalent to the algorithm in the flowchart?

(A) `include ← (floor > 10) OR (bedrooms = 3)`

(B) `include ← (floor > 10) AND (bedrooms = 3)`

(C) `include ← (floor ≤ 10) OR (bedrooms = 3)`

(D) `include ← (floor ≤ 10) AND (bedrooms = 3)`

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34. A researcher wrote a program to simulate the number of mice in an environment that contains predators. The program uses the following procedures.

Procedure Call	Explanation
`InitialMousePopulation ()`	Returns the number of mice at the start of the simulation
`InitialPredatorPopulation ()`	Returns the number of predators at the start of the simulation
`NextDayPopulation (numberOfMice, numberOfPredators)`	Based on the current numbers of mice and predators, returns the number of mice after one day

Code for the simulation is shown below.

```
days ← 0
numMice ← InitialMousePopulation ()
numPredators ← InitialPredatorPopulation ()
REPEAT UNTIL (days = 365)
{
   numMice ← NextDayPopulation (numMice, numPredators)
   days ← days + 1
}
DISPLAY ("There are")
DISPLAY (numMice)
DISPLAY ("mice after one year.")
```

Based on the code, which of the following assumptions is made in the simulation?

(A) The number of mice increases by 1 each day.

(B) The number of mice does not change from day to day.

(C) The number of predators increases by 1 each day.

(D) The number of predators does not change from day to day.

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35. A musician is creating a song using audio samples. Which of the following actions will minimize the risk of a copyright violation when creating sample-based music?

(A) Using samples found on popular music-streaming sites

(B) Using samples found on peer-to-peer networks

(C) Using samples from nondigital sound sources (vinyl records, tapes, etc.)

(D) Using samples published with a no-rights-reserved Creative Commons license

---

36. A computer program performs the operation `2 ÷ 3` and represents the result as the value `0.6666667`. Which of the following best explains this result?

(A) An overflow error occurred.

(B) The precision of the result is limited due to the constraints of using a floating-point representation.

(C) The program attempted to execute the operation with the arguments in reverse order.

(D) The program attempted to represent a floating-point number as an integer.

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37. The two code segments below are each intended to display the average of the numbers in the list `numList`. Assume that `numList` contains more than one value.

Program I:

```
sum ← 0
FOR EACH num IN numList
    sum ← sum + num
    avg ← sum / LENGTH numList
DISPLAY avg
```

Program II:

```
sum ← 0
FOR EACH num IN numList
    sum ← sum + num
avg ← sum / LENGTH numList
DISPLAY avg
```

Which of the following best describes the two code segments?

(A) Code segment I displays the correct average, but code segment II does not.

(B) Code segment II displays the correct average, but code segment I does not.

(C) Both code segments display the correct average, but code segment I requires more arithmetic operations than code segment II.

(D) Both code segments display the correct average, but code segment II requires more arithmetic operations than code segment I.

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38. A snack bar has a frequent customer program in which every 10th purchase is free. Customers are enrolled in the program when they make their first purchase. A programmer is writing a program to implement the frequent customer program. In one code segment, `cost` is set to `0` for every 10th purchase by a given customer. The programmer will use the procedure `GetCount (customerID)`, which returns the total number of purchases a customer has made since enrolling in the frequent customer program, including his or her first purchase.

Which of the following code segments will set `cost` to `0` for every 10th purchase a customer makes after enrolling in the frequent customer program?

(A)
```
count ← GetCount (customerID)
IF (count / 10 = 0)
{
   cost ← 0
}
```

(B)
```
count ← GetCount (customerID)
IF (NOT (count / 10 = 0))
{
   cost ← 0
}
```

(C)
```
count ← GetCount (customerID)
IF (count MOD 10 = 0)
{
   cost ← 0
}
```

(D)
```
count ← GetCount (customerID)
IF (NOT (count MOD 10 = 0))
{
   cost ← 0
}
```

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39. Which of the following explains a benefit of using open standards and protocols for Internet communication?

(A) Open standards and protocols allow different manufacturers and developers to build hardware and software that can communicate with hardware and software on the rest of the network.

(B) Open standards and protocols provide ways for users to eliminate the latency of messages they send on the Internet.

(C) Open standards and protocols allow users to freely share or reuse material found on the Internet for noncommercial purposes.

(D) Open standards and protocols prevent developers from releasing software that contains errors.

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Questions 40–41 refer to the information below.

Grades in a computer science course are based on total points earned on a midterm exam and a final exam. The teacher provides a way for students to improve their course grades if they receive high scores on the final exam: if a student's final exam score is greater than the student's midterm exam score, the final exam score replaces the midterm exam score in the calculation of total points.

The table below shows two students' scores on the midterm and final exams and the calculated total points each student earns.

Student Name	Midterm Exam Score	Final Exam Score	Total Points Calculation
Khalil	90	80	90 + 80 = 170
Josefina	70	90	90 + 90 = 180

- Khalil does better on the midterm exam than on the final exam, so his original midterm and final exam scores are added to compute his total points.
- Josefina does better on the final exam than on the midterm exam, so her final exam score replaces her midterm exam score in the total points calculation.

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40. The teacher has data representing the scores of thousands of students. For each student, the data contain the student name, the midterm exam score, the final exam score, and the result of the total points calculation. Which of the following could be determined from the data?

I. The average total points earned per student

II. The average increase in total points per student as a result of the score replacement policy

III. The proportion of students who improved their total points as a result of the score replacement policy

A) III only

B) I and II only

C) I and III only

D) I, II, and III

---

41. A programmer is writing a procedure to calculate a student's final grade in the course using the score replacement policy described. The student's exam scores are stored in the variables `midtermExam` and `finalExam`. The procedure `Max (a, b)` returns the larger of `a` and `b`.

Which of the following could be used in the procedure to calculate a student's total points earned in the course and store the result in the variable `adjustedTotal`?

(A) `adjustedTotal ← Max (midtermExam, finalExam)`

(B) `adjustedTotal ← Max (midtermExam, finalExam) + finalExam`

(C) `adjustedTotal ← Max (midtermExam, finalExam) + midtermExam`

(D) `adjustedTotal ← 2 * Max (midtermExam, finalExam)`

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42. Internet protocol version 4 (IPv4) represents each IP address as a 32-bit binary number. Internet protocol version 6 (IPv6) represents each IP address as a 128-bit binary number. Which of the following best describes the result of using 128-bit addresses instead of 32-bit addresses?

(A) 4 times as many addresses are available.

(B) 96 times as many addresses are available.

(C) 2^4 times as many addresses are available.

(D) 2^{96} times as many addresses are available.

---

43. An online retailer uses an algorithm to sort a list of n items by price. The table below shows the approximate number of steps the algorithm takes to sort lists of different sizes.

Number of Items	Number of Steps
10	100
20	400
30	900
40	1,600
50	2,500
60	3,600

Based on the values in the table, which of the following best characterizes the algorithm for very large values of n ?

(A) The algorithm runs in reasonable time.

(B) The algorithm runs, but not in reasonable time.

(C) The algorithm attempts to solve an undecidable problem.

(D) The algorithm attempts to find an approximate solution whenever it fails to find an exact solution.

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44. A computer program uses 4 bits to represent nonnegative integers. Which of the following statements describe a possible result when the program uses this number representation?

I. The operation `4 + 8` will result in an overflow error.

II. The operation `7 + 10` will result in an overflow error.

III. The operation `12 + 3` will result in an overflow error.

A) I only

B) II only

C) II and III only

D) I, II, and III

---

45. A `NAND` gate is a type of logic gate that produces an output of `false` only when both of its two inputs are `true`. Otherwise, the gate produces an output of `true`. Which of the following Boolean expressions correctly models a `NAND` gate with inputs `P` and `Q` ?

(A) `(NOT P) AND (NOT Q)`

(B) `(NOT P) AND Q`

(C) `NOT (P AND Q)`

(D) `NOT (P OR Q)`

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46. A student wants to create an algorithm that can determine, given any program and program input, whether or not the program will go into an infinite loop for that input.

The problem the student is attempting to solve is considered an undecidable problem. Which of the following is true?

(A) It is possible to create an algorithm that will solve the problem for all programs and inputs, but the algorithm can only be implemented in a low-level programming language.

(B) It is possible to create an algorithm that will solve the problem for all programs and inputs, but the algorithm requires simultaneous execution on multiple CPUs.

(C) It is possible to create an algorithm that will solve the problem for all programs and inputs, but the algorithm will not run in reasonable time.

(D) It is not possible to create an algorithm that will solve the problem for all programs and inputs.

---

47. In public key cryptography, the sender uses the recipient's public key to encrypt a message. Which of the following is needed to decrypt the message?

(A) The sender's public key

(B) The sender's private key

(C) The recipient's public key

(D) The recipient's private key

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48. In a certain science experiment, 75 percent of trials are expected to be successful and 25 percent of trials are expected to be unsuccessful. The program below is intended to simulate the results of repeated trials of the experiment.

```
successful ← 0
unsuccessful ← 0
REPEAT 1000 TIMES
{
   IF (<MISSING CODE>)
   {
      successful ← successful + 1
   }
   ELSE
   {
      unsuccessful ← unsuccessful + 1
   }
}
DISPLAY (successful)
DISPLAY ("trials were successful,")
DISPLAY (unsuccessful)
DISPLAY ("trials were unsuccessful.")
```

Which of the following can be used to replace `<MISSING CODE>` so that the simulation works as intended?

(A) `RANDOM (1, 100) = 25`

(B) `RANDOM (1, 100) ≤ 25`

(C) `RANDOM (1, 100) = 75`

(D) `RANDOM (1, 100) ≤ 75`

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49. The code segment below is intended to display all multiples of 5 between the values `start` and `end`, inclusive. For example, if `start` has the value `35` and `end` has the value `50`, the code segment should display the values `35`, `40`, `45`, and `50`. Assume that `start` and `end` are multiples of 5 and that `start` is less than `end`.

```
Line 1:  i ← start
Line 2:  REPEAT <MISSING EXPRESSION> TIMES
Line 3:  {
Line 4:     DISPLAY (i)
Line 5:     i ← i + 5
Line 6:  }
```

Which of the following could replace `<MISSING EXPRESSION>` in line 2 so that the code segment works as intended?

(A) `end - start + 1`

(B) `end - start + 6`

(C) `((end - start) / 5) + 1`

(D) `5 * (end - start) + 1`

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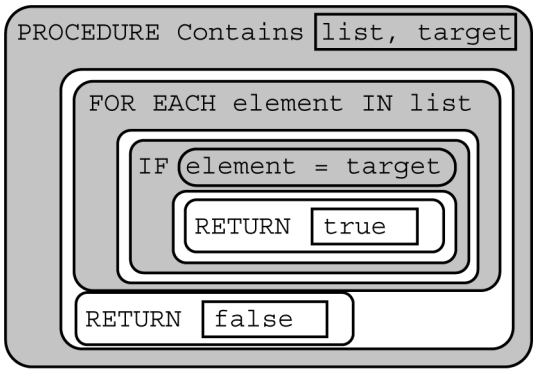
50. The procedure below searches for the value `target` in `list`. It returns `true` if `target` is found and returns `false` otherwise.



Which of the following are true statements about the procedure?

I. It implements a binary search.

II. It implements a linear search.

III. It only works as intended when `list` is sorted.

(A) I only

(B) II only

(C) I and III

(D) II and III

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51. Which of the following is an example of symmetric encryption?

(A) Evy buys a locked box that operates using two different codes. When the first code is entered, a slot opens that allows a message to be put in the box. When the second code is entered, the door to the box opens. Evy gives the first code to her friends so they can leave messages for her and keeps the second code to herself so that she is the only one who can retrieve the messages.

(B) Finn and Gwen develop a system that maps each letter of the alphabet to a unique symbol using a secret key. Finn uses the key to write a message to Gwen where each letter is replaced with the corresponding symbol. Gwen uses the key to map each symbol back to the original letter.

(C) Hannah writes a message to send to Isabel and hides the message under a rock behind the soccer field. Hannah gives Isabel the exact location of the rock so that only Isabel can find the message.

(D) Juan writes a message to send to Kelly and slides the message through a slot in the front of Kelly's locker. Juan knows that Kelly has not shared her locker combination with anyone, so no one other than Kelly will be able to read the message.

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52. A programmer is developing a word game. The programmer wants to create an algorithm that will take a list of words and return a list containing the first letter of all words that are palindromes (words that read the same backward or forward). The returned list should be in alphabetical order. For example, if the list contains the words `["banana", "kayak", "mom", "apple", "level"]`, the returned list would contain `["k", "l", "m"]` (because `"kayak"`, `"level"`, and `"mom"` are palindromes).

The programmer knows that the following steps are necessary for the algorithm but is not sure in which order they should be executed.

Step	Explanation
Shorten	Takes a list of words and returns a new list that contains only the first letter of each word from the input list
Keep palindromes	Takes a list of words and returns a list that contains only the palindromes from the input list
Sort	Takes a list of words and returns a copy of the list in alphabetical order

Executing which of the following sequences of steps will enable the algorithm to work as intended?

I. First shorten, then keep palindromes, then sort

II. First keep palindromes, then shorten, then sort

III. First sort, then keep palindromes, then shorten

(A) I only

(B) II only

(C) I and III

(D) II and III

---

53. Which of the following best describes the purpose of machine learning programs?

(A) To analyze large data sets, recognize patterns, and make predictions based on data

(B) To automatically translate algorithms from natural language to machine language

(C) To determine whether an algorithm can be constructed to answer “yes” or “no” for all possible inputs

(D) To find approximate solutions to problems that would otherwise require an unreasonably long amount of time to solve

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54. A programmer notices the following two procedures in a library. The procedures do similar, but not identical, things.

- Procedure `Square (n)` returns the value n^2.
- Procedure `Cube (n)` returns the value n^3.

Which of the following procedures is a generalization of the procedures described above?

(A) Procedure `Add (n, m)`, which returns the value $n + m$

(B) Procedure `Fourth (n)`, which returns the value n^4

(C) Procedure `Polynomial(n)`, which returns the value $n^3 + n^2$

(D) Procedure `Power (n, m)`, which returns the value n^m

---

55. A student wrote the procedure below, which is intended to ask whether a user wants to keep playing a game. The procedure does not work as intended.

```
PROCEDURE KeepPlaying ()
{
   DISPLAY ("Do you want to continue playing (y/n)?")
   response ← INPUT ()
   IF ((response = "y") AND (response = "yes"))
   {
      RETURN (true)
   }
   ELSE
   {
      RETURN (false)
   }
}
```

Which of the following best describes the result of running the procedure?

(A) The procedure returns `true` when the user inputs the value `"y"` and returns `false` otherwise.

(B) The procedure returns `true` when the user inputs the value `"n"` and returns `false` otherwise.

(C) The procedure returns `true` no matter what the input value is.

(D) The procedure returns `false` no matter what the input value is.

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56. Consider the following program.

```
count ← 1
sum ← 0
REPEAT 10 TIMES
    sum ← sum + count
    count ← count + 2
DISPLAY [sum]
```

Which of the following describes the result of executing the program?

(A) The program displays the sum of the even integers from 0 to 10.

(B) The program displays the sum of the even integers from 0 to 20.

(C) The program displays the sum of the odd integers from 1 to 9.

(D) The program displays the sum of the odd integers from 1 to 19.

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57. A program contains the following procedures for string manipulation.

Procedure Call	Explanation
`Concat (str1, str2)`	Returns a single string consisting of `str1` followed by `str2`. For example, `Concat("key", "board")` returns `"keyboard"`.
`Substring (str, start, length)`	Returns a substring of consecutive characters from `str`, starting with the character at position `start` and containing `length` characters. The first character of `str` is located at position `1`. For example, `Substring ("delivery", 3, 4)` returns `"live"`.

Which of the following expressions can be used to generate the string `"Happy"` ?

(A) `Concat (Substring ("Harp", 1, 1), Substring ("Puppy", 2, 4))`

(B) `Concat (Substring ("Harp", 1, 2), Substring ("Puppy", 3, 3))`

(C) `Concat (Substring ("Harp", 1, 2), Substring ("Puppy", 4, 2))`

(D) `Concat (Substring ("Harp", 2, 2), Substring ("Puppy", 4, 2))`

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58. Which of the following are true statements about how the Internet enables crowdsourcing?

I. The Internet can provide crowdsourcing participants access to useful tools, information, and professional knowledge.

II. The speed and reach of the Internet can lower geographic barriers, allowing individuals from different locations to contribute to projects.

III. Using the Internet to distribute solutions across many users allows all computational problems to be solved in reasonable time, even for very large input sizes.

A) I and II only

B) I and III only

C) II and III only

D) I, II, and III

---

59. A media librarian at a movie studio is planning to save digital video files for archival purposes. The movie studio would like to be able to access full-quality videos if they are needed for future projects. Which of the following actions is LEAST likely to support the studio's goal?

(A) Using video file formats that conform to published standards and are supported across many different devices

(B) Using lossy compression software to reduce the size requirements of the data being stored

(C) Using storage media that can be expanded for additional data capacity

(D) Using a system that incorporates redundancy to handle disk failure

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Directions: For each of the questions or incomplete statements below, two of the suggested answers are correct. For each of these questions, you must select both correct choices to earn credit. No partial credit will be earned if only one correct choice is selected. Select the two that are best in each case and then fill in the corresponding circles that begin with number 131 on the answer sheet.

---

131. Which of the following are ways in which a programmer can use abstraction to manage the complexity of a program? Select two answers.

(A) Replacing each instance of repeated code with a call to a procedure

(B) Replacing longer variable names with shorter variable names to reduce typing errors

(C) Replacing several lines of documentation with a single line of documentation

(D) Replacing the string variables `name1`, `name2`, `name3`, and `name4` with a list of strings called `names`

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132. Two different schools maintain data sets about their currently enrolled students. No individual student is enrolled at both schools. Each line of data contains information, separated by commas, about one student.

East High School stores the data in the following format.

Data format	**School Name, Last Name, First Name, Age, ZIP Code, Days Absent**
Sample line of data	`East H.S., Ableson, Carlton, 16, 98987, 8`

West High School stores the data in the following format.

Data format	**Student ID, First Name, Last Name, Age, Days Absent**
Sample line of data	`87667721, Kendra, Angelos, 14, 12`

The two schools would like to combine their data to make a single data set. Which of the following can be done with the combined data?

Select two answers.

(A) The schools can create a single list of student names, sorted by last name.

(B) The schools can determine the average number of days students are absent.

(C) The schools can determine which ZIP code is represented by the most students.

(D) The schools can determine the student ID of the student with the greatest number of absences.

---

133. A free online encyclopedia contains articles that can be written and edited by any user. Which of the following are advantages the online encyclopedia has over a traditional paper-based encyclopedia?

Select two answers.

(A) The ability to easily check that the encyclopedia is free of copyrighted content

(B) The ability to ensure that encyclopedia content is the same every time it is accessed

(C) The ability to have a larger number of perspectives reflected in the encyclopedia content

(D) The ability to quickly update encyclopedia content as new information becomes available

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134. The program below is intended to count the number of prime numbers in a list called `numbers` and display the result. The program uses the procedure `isPrime (n)`, which returns `true` if `n` is a prime number and `false` otherwise.

The program does not work as intended.

```
Line 1:  count ← 0
Line 2:  FOR EACH value IN numbers
Line 3:  {
Line 4:     count ← 0
Line 5:     IF (isPrime (value))
Line 6:     {
Line 7:        count ← count + 1
Line 8:     }
Line 9:     count ← count + 1
Line 10: }
Line 11: DISPLAY (count)
```

Which two lines of code should be removed so that the program will work as intended?

Select two answers.

(A) Line 1

(B) Line 4

(C) Line 7

(D) Line 9

---

135. Which of the following statements describe how cloud computing has affected Internet communication?

Select two answers.

(A) Cloud computing has eliminated the need to provide redundancy in Internet routing.

(B) Cloud computing has helped enhance collaboration.

(C) Cloud computing has introduced new data-security concerns.

(D) Cloud computing has reduced concerns about intellectual property rights.

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136. A program is created to perform arithmetic operations on positive and negative integers. The program contains the following incorrect procedure, which is intended to return the product of the integers `x` and `y`.

```
PROCEDURE Multiply (x, y)
{
   count ← 0
   result ← 0
   REPEAT UNTIL (count = y)
   {
      result ← result + x
      count ← count + 1
   }
   RETURN (result)
}
```

A programmer suspects that an error in the program is caused by this procedure. Under which of the following conditions will the procedure NOT return the correct product?

Select two answers.

(A) When the values of `x` and `y` are both positive.

(B) When the value of `x` is positive and the value of `y` is negative.

(C) When the value of `x` is negative and the value of `y` is positive.

(D) When the values of `x` and `y` are both negative.

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137. The procedure `Smallest` is intended to return the least value in the list `numbers`. The procedure does not work as intended.

```
PROCEDURE Smallest (numbers)
{
   min ← numbers[1]
   FOR EACH number IN numbers
   {
      IF (number < min)
      {
         RETURN (number)
      }
   }
   RETURN (min)
}
```

For which of the following values of `theList` will `Smallest (theList)` NOT return the intended value?

Select two answers.

(A) `theList ← [10, 20, 30, 40]`

(B) `theList ← [20, 10, 30, 40]`

(C) `theList ← [30, 40, 20, 10]`

(D) `theList ← [40, 30, 20, 10]`

STOP

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS TEST.